KARTIK KARKERA

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Education

University of Texas, Dallas — Dallas, TX

Master of Science in Computer Science - GPA: 3.67 / 4.0 — *May 2026*

- **Relevant Coursework:** Distributed Systems, Database Systems, Machine Learning

Technical Skills

Backend & APIs: FastAPI, Flask, GraphQL
Cloud & Infra: Amazon Web Services, Docker, Linux, GitHub Actions (CI/CD)
Data & Streaming: Apache Spark, Kafka, Databricks, Pandas
Databases & Search: PostgreSQL, DynamoDB, Snowflake, Elasticsearch, MongoDB, Redis
AI & Agentic: LangChain, Bedrock, RAG, Vector Databases

Certifications

Databricks Certified Associate Developer for Apache Spark - Python — Jan 2026
AWS Certified Solutions Architect – Associate (SAA-C03) — Jul 2025
AWS Certified Data Engineer – Associate (DEA-C01) — Sept 2025

Experience

University of Texas, Dallas — Richardson, TX

Computer Science Teaching Assistant — *Aug 2024 – May 2026*

- Helped 90+ students across 2 semesters work through Computer Architecture concepts - MIPS assembly, memory hierarchy, and low-level programming, earning consistent positive feedback from the course professor.
- Graded assignments and ran project presentations, giving students detailed technical feedback on their implementations.

PyCray Technologies — Noida, India

Software Developer — *May 2023 – Jun 2024*

- Built a serverless, event-driven pipeline for an algorithmic trading platform using DynamoDB Streams and AppSync, broadcasting PnL updates on every position change with <50ms end-to-end latency.
- Architected a hot-cold storage strategy (DynamoDB for active trades, PostgreSQL for historical analytics), decoupling query paths to maintain performance at scale across both workloads.
- Implemented EventBridge-triggered Lambda pipelines to migrate data from DynamoDB to PostgreSQL, handling schema evolution and cross-store consistency.
- Optimized ride-search on a carpooling platform using geospatial Elasticsearch indexing, reducing location-based query latency.
- Developed and optimized GraphQL APIs integrated with ReactJS dashboards, unifying 3+ data sources into real-time reporting views for internal analytics.

Trivia Softwares — Mumbai, India

Software Developer Intern — *Aug 2021 – Sep 2021*

- Built a Django REST API with secure user authentication and email verification, improving onboarding security and reducing manual account management overhead.

Projects

Traceline - Pipeline Anomaly Detection | *Python, Flask, Pandas, NumPy, React* | **1st Place - TidalHack @ TAMU**

- Designed an ingestion and alignment pipeline processing 15+ years of inspection records for RCP Inc., correlating multi-run scan data to track corrosion growth over time and predict structural failure risks.
- Engineered a relative-matching algorithm using fixed hardware anchors to correct odometer drift across inspection runs, enabling accurate anomaly co-registration across datasets spanning different scan sessions.
- Built a React-based 3D visualization dashboard enabling field engineers to interactively track anomaly evolution across pipeline

Data Centric Hackathon | *Python, PyTorch, 3LC, AWS* | **3rd Place - 3LC × AWS Hackathon @ UTD**

- Optimized a ResNet-18 image classifier under strict constraints - no architecture changes, no external data, no pretrained weights - pushing validation accuracy from 70% to 94% purely through data quality improvements.
- Used 3LC Dashboard to iteratively relabel low-confidence samples, rebalance class distributions, and tune per-sample loss weights derived from embedding cluster analysis across training runs.